# Nowcasting European LNG send-out from free public data: measured revision processes, calibrated filtering, and a physical stress index

Solomon J. Williams
University of Edinburgh

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**Abstract**

[**TODO-SOLOMON:** 150 words. Claims to compress: (i) commercial-grade LNG supply signal rebuilt entirely from free public sources, with the two layers vendors do not publish — the revision process of the official data and calibrated uncertainty; (ii) a semi-Markov, Rao-Blackwellized particle filter whose evening nowcasts are 3–7× more accurate than any daily baseline with distribution-free coverage guarantees; (iii) empirical findings en route: official nominations are worse than persistence, the official intraday estimate appears to be a carry-forward, intraday deviations are one random ramp (fBm bridge $\hat{H} \approx 0.98$), arrivals are emphatically non-Poisson; (iv) a days-of-cover stress index that adds AUC over storage fullness in-sample, presented with its caveats.]

## 1 Introduction

[**TODO-SOLOMON:** 2 pages. Outline: LNG send-out as a price-relevant hidden variable; vendors sell it from proprietary satellite AIS; thesis = the open reconstruction plus what vendors skip (revisions, calibration); contributions list mirroring the abstract; related work in three strands — vessel-ETA prediction (statistical/ML taxonomies, no stochastic-process treatment), maritime inventory routing (prescriptive, import-terminal inventories under-modelled, uncertainty "in its infancy"), LNG arbitrage economics (damped spread response). Cite the deep-research-verified sources.]

## 2 Data: five free sources and their measured behaviour

[**TODO-SOLOMON:** Prose around Table **??**; the measured-timing findings are the section's substance — each was verified live and several contradict the platforms' documentation.]

## 3 The revision processes of the official data

### 3.1 The UK ladder, reconstructed retroactively

[**TODO-SOLOMON:** Prose: latestFlag=N serves every historical published version. Findings: 27.8% of values republished at ∼30 d with median change zero; heavy tail to 662.7 GWh (send-out) and 340 GWh (stock, next-day); the "M+15" rung publishes at ∼M+1 and never differs

| Source | Access | Cadence | Measured behaviour |
|---|---|---|---|
| ENTSOG transparency | keyless | daily/hourly | daily D $\rightarrow$ 09:20 D+1; hourly $\sim$2 h lag; nominations for D at 16:00 D$-$1; UK rows backfilled $\sim$6 d; some queries hang with zero bytes |
| GIE ALSI | free key | daily | 19:30 + 23:00 publications; dual-unit inventory (V13); UK absent since $\geq$2024; intraday E-row = previous day's C (carry-forward; preliminary) |
| National Gas portal | keyless | daily + 2-min | D+1 physical at 12:01; D+2/M+15 ladder; per-sub-terminal 2-minutely instantaneous flows (not archived by the operator) |
| aisstream.io AIS | free key | streaming | terrestrial only; destination fields $\sim$52% erroneous (lit.); berth calls resolvable; receiver gaps at Milford Haven/Dunkerque |
| AGSI / prices | free key / open | daily | EU storage aggregate; UK SAP; (TTF via Yahoo, analysis only) |

Table 1: Sources and live-measured timing. [**TODO-SOLOMON:** tighten wording; add access dates.]

from final D+2 (max $\Delta = 0.00$ over 2,829 matched days); D+1 physical $\approx$ final on 99%+ of days (p99 0.02 GWh). Showcase: Grain 2025-04-21 — physical and ENTSOG record $\sim$662 GWh, commercial retracted to 0 a month later; the official record permanently self-contradicts, and that day's D+1 published three days late.]

### 3.2 The EU intraday estimate

[**TODO-SOLOMON:** Carry-forward finding with the snapshot-archive methodology; state the evidence window honestly (n days at time of writing) and the implication: the official platform's intraday row carries no information, so an ENTSOG-hourly nowcast is ahead by construction.]

## 4 Cargo arrivals from mass balance

[**TODO-SOLOMON:** Derive $A_t = \Delta I_t + S_t$; describe clustering; then:] [**TODO-SOLOMON:** Renewal structure: empirical hazard is exactly zero at 1-day gaps at every terminal (berth turnaround); rising hazards at scheduled terminals (Gate 0.56 by day 8). Dragon's staged discharges; multi-berth chaining. Connect to the U6 arrival-hazard filter input.]

| | median event (GWh) | 172k-class reference | deviation |
|---|---|---|---|
| Dunkerque | 1095 | 1096 | −0.1% |
| Gate | 1105 | 1096 | +0.8% |
| Zeebrugge | 1096 | 1096 | 0.0% |
| Montoir | 1092 | 1096 | −0.4% |

Table 2: 1,446 extracted arrivals; EU medians vs. the vessel-class discharge energy implied by independent physics (capacity × fill × boil-off × heel × GCV).

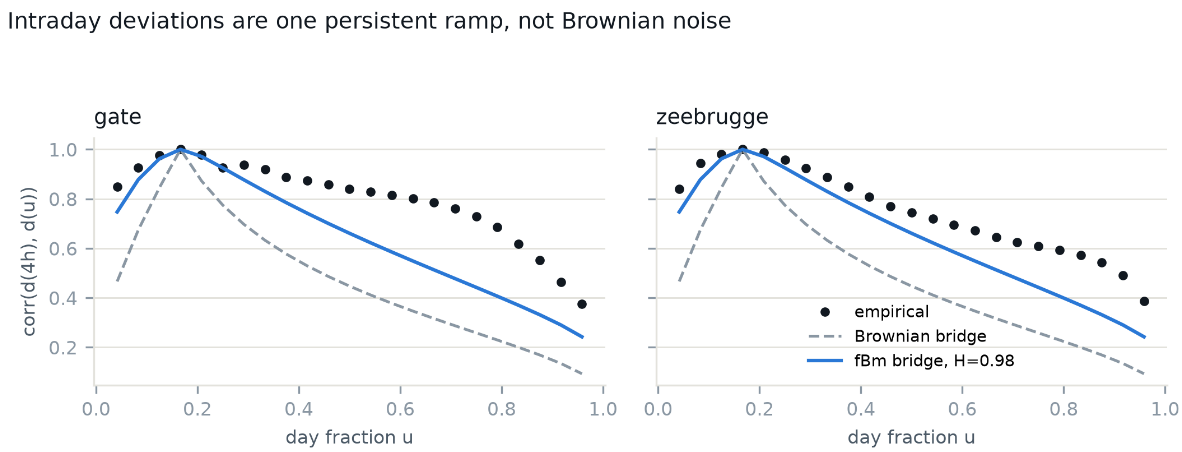


Figure 1: Correlation of intraday deviations $d(u)$ with the hour-4 value: the empirical structure (dots) tracks a fractional-Brownian bridge with $H = 0.98$, far from Brownian. [**TODO-SOLOMON:** caption polish.]

## 5 A semi-Markov, Rao-Blackwellized nowcaster

[**TODO-SOLOMON:** The model section — yours entirely. Content that must appear: (i) explicit-duration regimes with NB sojourns as first-passage laws (motivate hazard shapes from noise-induced escape); (ii) conditional linear-Gaussianity and the marginalized filter, idle as the exact $S = 0$ boundary state; (iii) the bridge observation model for intraday cumulatives and the empirical kernel finding: fBm-bridge $\hat{H} \approx 0.98$ at all hourly terminals ($-81$ to $-87\%$ correlation misfit vs Brownian) — one random ramp factor per day; (iv) PGAS parameter posteriors, the identifiability lessons (prior-tethering of unidentified sojourn blocks; mixture label ordering), and parameter uncertainty in the bands.]

## 6 Walk-forward evaluation

[**TODO-SOLOMON:** Discussion: H0 ties persistence *because* nominations are empirically worse than persistence and are rejected by the fitted weight — a data finding, not a modelling failure. Calibration: raw vs conformal, Gibbs–Candès guarantee and its marginal-only nature.]


Figure 2: Gate, June–August 2026: the evening nowcast (with 90% band) against the figure published the next morning; the band widens exactly at the outage cliffs. **[TODO-SOLOMON:** caption polish.]

| Terminal | H0 | H1 (midday) | H2 (evening) | persistence | nominations | cov90 (conf.) |
|---|---|---|---|---|---|---|
| Gate | 20.7 | 8.4 | **4.5** | 20.6 | 63.0 | 0.92 |
| Eems | 15.3 | 4.4 | **2.1** | 15.1 | 35.8 | 0.93 |
| Zeebrugge | 47.1 | 32.0 | **24.3** | 47.1 | 120.3 | 0.87 |
| Dunkerque | 29.9 | – | – | 29.8 | 225.1 | 0.90 |
| Grain | 19.6 | – | – | 19.7 | – | 0.89 |
| South Hook | 18.1 | – | – | 17.6 | – | 0.91 |

Table 3: MAE (GWh/d), 821 walk-forward days per terminal, 2024-06–2026-08; bootstrap reference filter with bridge observations and adaptive-conformal bands. **[TODO-SOLOMON:** add the RBPF column (500 particles, 15× faster, comparable intraday, $p$(idle) output); describe horizons and the leak-freedom of every fit.]

## 7 A physical stress index

**[TODO-SOLOMON:** Define days-of-cover; report the in-sample horse race — storage-% alone AUC 0.77–0.80 on TTF q97.5 spikes, +cover 0.85 (SAP: 0.62 → 0.73) — and give the caveats their own paragraph: in-sample thresholds, overlapping windows, two winters, seasonal confounding signatures in two components; specify the pre-registered out-of-sample design for winter 2026–27 (storage entered September at 65% vs ∼90% typical — the test will run live). Ship-pipeline layer: departures/chokepoints/berths sampling design and why mid-ocean satellite AIS is out of scope.]

## 8 Economic content: a bounded null

**[TODO-SOLOMON:** The event study: ordinary daily surprises ($\sigma \approx 109$ GWh) move front-month TTF $\leq 0.16\%$ same-day, $|t| < 1$ over 564 trading days — a bound, consistent with

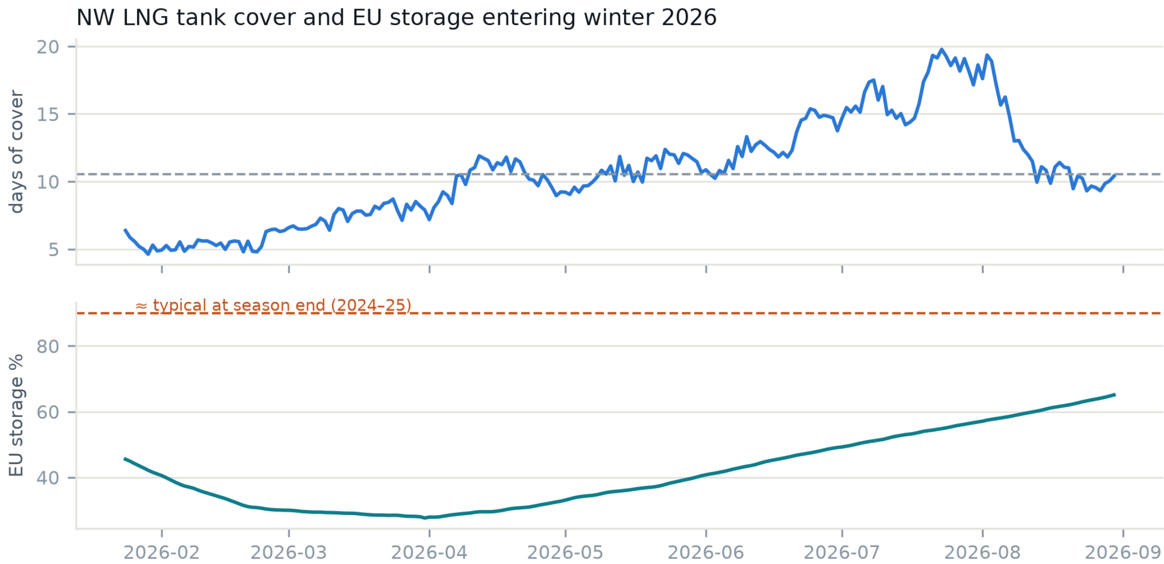


Figure 3: Days-of-cover and EU storage fullness entering winter 2026. [**TODO-SOLOMON:** caption polish.]

damped-arbitrage structure; implications for where value can live (prompt products, tails, conditional windows).]

# 9 Limitations

[**TODO-SOLOMON:** Own every weakness first: no historical satellite AIS (forward-only ship layer); Zeebrugge residual miscalibration; cov50 width; UK intraday horizons forward-only; stress study not yet out-of-sample; monetization honesty (ICE TTF granularity vs a student account).]

# Reproducibility

All code, the snapshot archives, and every table's generating script are at `https://github.com/solwilliams6305/LNG-send-out-nowcasting`. Data sources are credited per their terms (GIE ALSI/AGSI as data source; ENTSOG; National Gas; aisstream.io). [**TODO-SOLOMON:** pin commit hash at submission.]

# References

[**TODO-SOLOMON:** Populate from the memo's reference list: Gibbs & Candès 2021/2024; Zaffran et al. 2022; Schön–Gustafsson–Nordlund 2005; Lindsten–Jordan– Schön 2014; Yu 2010 + JRSS-C 2025 zero-inflated HSMM; the AIS destination-quality papers; the MIRP survey; LNG arbitrage economics.]